STPM 2017 PHYSICS 960/3

Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ *J s,* $c = 3.00 \times 10^{8}$ *m* s^{-1}, $e = 1.60 \times 10^{-19}$ *C,* $m_e = 9.11 \times 10^{-31}$ *kg, 1 u* $= 1.66 \times 10^{-27}$ *kg.*

Section A [15 marks]

1. A particle performs a simple harmonic motion with amplitude, A. What is the maximum velocity of the particle to complete N oscillations in time, t?

A $\frac{4\pi^2N^2A}{t^2}$ B $\frac{4\pi^2t^2A}{N^2}$ C $\frac{2\pi NA}{t}$ D $\frac{2\pi tA}{N}$

Answer: C

$$f = \frac{N}{t} \Longrightarrow \omega = \frac{2\pi N}{t};\ v_{\max} = A\omega = \frac{2\pi NA}{t}$$

Explanation: The frequency is the number of oscillations per unit time.

2. The bob of a simple pendulum is displaced and then released. The displacement, x, of the bob after time, t, is given by $x = 0.50 \cos 2\pi t$. What is the shortest time when the kinetic energy of the system is equal to its potential energy?

A 0.13 s B 0.17 s C 0.25 s D 0.38 s

Answer: A

$E_k = E_p$ when each is half the total energy: $\frac{1}{2}kx^2 = \frac{1}{4}kA^2 \Longrightarrow x = A/\sqrt{2}$

$$\cos 2\pi t = \frac{1}{\sqrt{2}} \Longrightarrow 2\pi t = \frac{\pi}{4} \Longrightarrow t = 0.125 \text{ s} \approx 0.13 \text{ s}$$

Explanation: The bob starts at maximum displacement (cos term), where all the energy is potential.

3. A transmitter radiates energy uniformly in all directions. At a distance of 50 km from the source, the amplitude of vibration of air molecules is 7.98×10^{-3} m. If there is no energy loss to the surroundings, what is the amplitude of the vibration at a distance of 30 km from the source?

A 2.9×10^{-3} m B 4.8×10^{-3} m C 1.3×10^{-2} m D 2.2×10^{-2} m

Answer: C

$$I \propto A^2 \text{ and } I \propto 1/r^2 \Longrightarrow \text{amplitude} \propto 1/r$$

$$A_2 = (7.98 \times 10^{-3})\frac{50}{30} = 1.3 \times 10^{-2} \text{ m}$$

Explanation: Closer to the source the amplitude is larger.

4. Which statement is true about standing wave?

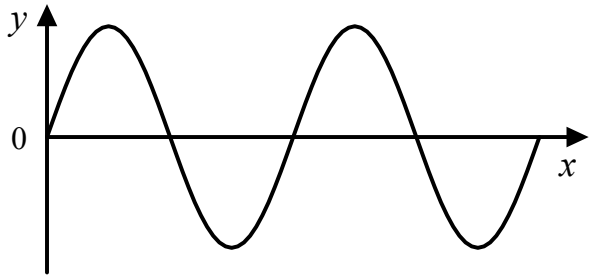
A Points on the nodes have maximum amplitude.
B Points between two adjacent nodes are in phase.
C Every point on the waves has the same amplitude.
D Points on the antinodes have the highest frequency.

Answer: B

All points between two adjacent nodes oscillate in phase.

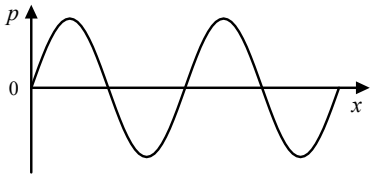
Explanation: Nodes have zero amplitude; amplitude varies along the wave; every point oscillates at the same frequency.

5. The propagation of sound waves in air in the positive-x direction from a source is represented by the graph of displacement, y, against distance, x, below.

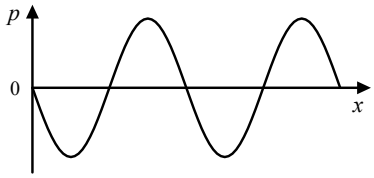


Which graph shows the variation of air pressure, p, with distance, x, from the source?

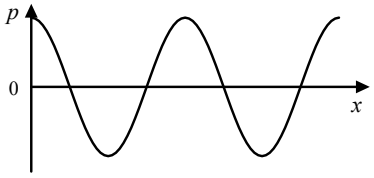
A



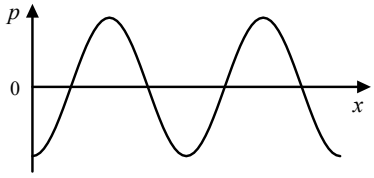
B



C



D



Answer: D

$$p \propto -\frac{\partial y}{\partial x}\text{: for } y = y_0 \sin kx,\ p \propto -\cos kx$$

The pressure variation starts at its negative maximum and is 90° out of phase with the displacement (D).

Explanation: Compressions (maximum pressure) occur where the displacement is zero and the neighbouring particles move towards each other.

6. An elastic string is stretched between two fixed points which are separated at 2.6 m apart. If the mass of the string is 0.15 kg and the tension in the string is 30 N, what is the fundamental frequency of vibration that can be produced by the string?

A 2.7 Hz B 4.4 Hz C 5.4 Hz D 8.8 Hz

Answer: B

$$\mu = \frac{0.15}{2.6} = 0.0577 \text{ kg m}^{-1};\ v = \sqrt{\left(\frac{30}{0.0577}\right)} = 22.8 \text{ m s}^{-1}$$

$$f = \frac{v}{2L} = \frac{22.8}{2(2.6)} = 4.4 \text{ Hz}$$

Explanation: The fundamental has one loop: $\lambda = 2L$.

7. The radius of curvature of a spherical concave mirror is 30 cm. What are the characteristics of the image formed by the mirror if an object is placed 35 cm in front of the mirror?

A The image is upright, real, magnified and located behind the mirror.
B The image is upright, real, diminished and located behind the mirror.
C The image is inverted, real, magnified and located in front of the mirror.
D The image is inverted, real, diminished and located in front of the mirror.

Answer: D

$f = 15$ cm; the object (35 cm) is beyond the centre of curvature (30 cm).
Image: real, inverted, diminished, between F and C in front of the mirror.

Explanation: Check: $1/v = 1/15 - 1/35 \Longrightarrow v = 26.3$ cm, $m = 0.75$.

8. A very thin transparent coating of refractive index 1.25 is deposited on a lens of refractive index 1.50 to reduce reflection. What is the minimum thickness of the coating so that normal incident light of wavelength 550 nm undergoes destructive interference after reflection from the lens?

A 5.5×10^{-8} m B 1.1×10^{-7} m C 1.4×10^{-7} m D 1.8×10^{-7} m

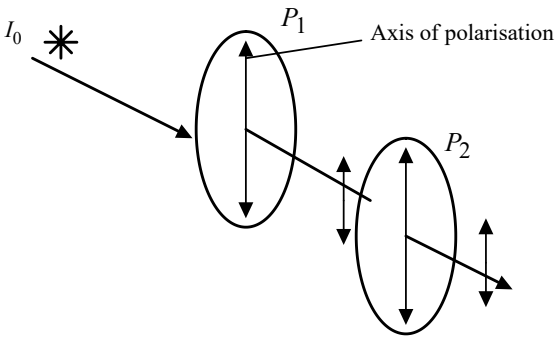
Answer: B

Both reflections (air → coating and coating → glass) have a phase change of π, so they cancel.

$$\text{Destructive: } 2nt = \frac{\lambda}{2} \Longrightarrow t = \frac{\lambda}{4n} = \frac{550 \times 10^{-9}}{4(1.25)} = 1.1 \times 10^{-7} \text{ m}$$

Explanation: This is a non-reflecting (anti-reflection) coating, a quarter wavelength thick in the coating.

9. Two polarisers, P_1 and P_2, with their axis of polarisation parallel to each other are shown in the diagram below.



P_1 is illuminated with unpolarised light of intensity, I_0. Through what angle must P_2 be rotated so that the intensity of light emerging from P_2 becomes $0.22I_0$?

A 48° B 62° C 64° D 77°

Answer: A

$$\text{After } P_1\text{: } \frac{I_0}{2}\text{; after } P_2\text{: } \frac{I_0}{2}\cos^2\theta = 0.22I_0$$

$$\cos^2\theta = 0.44 \Longrightarrow \theta = 48°$$

Explanation: The first polariser halves the intensity of unpolarised light; Malus's law applies at the second.

10. An optical fiber is a physical structure usually made of plastic or glass which is able to transmit light through it. Which statement is **not** true about the optical fibers?

A Optical fibers are used in telecommunication and medical diagnostic.
B Optical fibers use total internal reflection to transmit light through it.
C Optical fibers cause almost no loss in intensity when light is transmitted through it.
D Optical fibers consists of a transparent medium surrounded by a material of higher refractive index.

Answer: D

The core must have a **higher** refractive index than the surrounding cladding for total internal reflection. Statement D is not true.

Explanation: Optical fibres are used in telecommunications and medicine and have very low loss.

11. In a photoelectric effect experiment using monochromatic light, some of the electrons emitted have a maximum kinetic energy because

A the incident photons have the same energy
B the photoelectric emission is a random process
C the energy of photon is absorbed by an electron
D the electrons escape from the surface with minimum work

Answer: D

The electrons with the maximum kinetic energy are those that need the least energy to escape, i.e. electrons at the surface, which escape doing only the work function.

Explanation: Electrons deeper inside lose extra energy in collisions before escaping, so they emerge with less kinetic energy.

12. X-ray beam of wavelength 8.40×10^{-10} m is incident on a crystal. The third order diffraction of the X-ray occurs at a glancing angle of 24.5°. What is the separation of the two adjacent atomic planes of the crystal?

A 1.38×10^{-9} m B 2.76×10^{-9} m C 3.04×10^{-9} m D 6.10×10^{-9} m

Answer: C

$$2d \sin\theta = 3\lambda \Longrightarrow d = \frac{3(8.40 \times 10^{-10})}{2 \sin 24.5°} = 3.04 \times 10^{-9} \text{ m}$$

Explanation: Use $n = 3$ in Bragg's law.

13. Which statement is **not** true about nanoscience?

A In nanoscience, electronic devices can be miniaturised.
B The study of structures and materials on the scale of 10^9 m.
C To study nanostructures, electron microscopes need to be used.
D The study of structures and materials on the scale of 1 to 100 nm.

Answer: B

Nanoscience deals with structures on the scale of 10^{-9} m (1–100 nm), not 10^9 m. Statement B is not true.

Explanation: Electron microscopes are needed because nanostructures are smaller than the wavelength of visible light.

14. What is the nuclear binding energy per nucleon for $^{16}_{8}$O atom?

[The mass of the oxygen atom is 2.65516×10^{-26} kg, mass of neutron is 1.67444×10^{-27} kg and mass of hydrogen atom is 1.67295×10^{-27} kg.]

A 1.28×10^{-12} J B 2.56×10^{-12} J C 1.31×10^{-10} J D 2.61×10^{-10} J

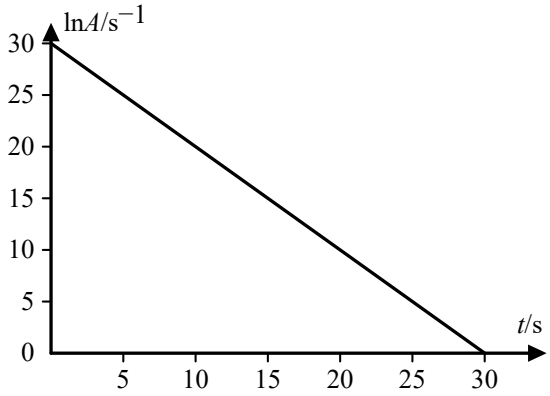
Answer: A

$$\Delta m = 8(1.67295 \times 10^{-27}) + 8(1.67444 \times 10^{-27}) - 2.65516 \times 10^{-26} = 2.275 \times 10^{-28} \text{ kg}$$

$$E_B/A = \frac{(2.275 \times 10^{-28})(3.00 \times 10^8)^2}{16} = 1.28 \times 10^{-12} \text{ J}$$

Explanation: Using hydrogen-atom masses (not proton masses) makes the electrons balance with the oxygen atom's 8 electrons.

15. The variation of $\ln A$ with time, t, for a radioactive sample where A is the rate of decay is shown in the graph below.



What is the half-life and the initial number of nuclei in the sample?

	Half-life	*Initial number of nuclei*
A	0.69 s	1.1×10^{13}
B	0.69 s	1.9×10^{14}
C	12.5 s	1.9×10^{14}
D	12.5 s	1.1×10^{13}

Answer: A

$$\ln A = \ln A_0 - \lambda t\text{: gradient} = -\frac{30}{30} \Longrightarrow \lambda = 1.0\ \text{s}^{-1},\ T_{½} = 0.69\ \text{s}$$

$$A_0 = \text{e}^{30} = 1.07 \times 10^{13}\ \text{s}^{-1};\ N_0 = \frac{A_0}{\lambda} = 1.1 \times 10^{13}$$

Explanation: The intercept gives $\ln A_0$; the magnitude of the gradient is the decay constant.

Section B [15 marks]

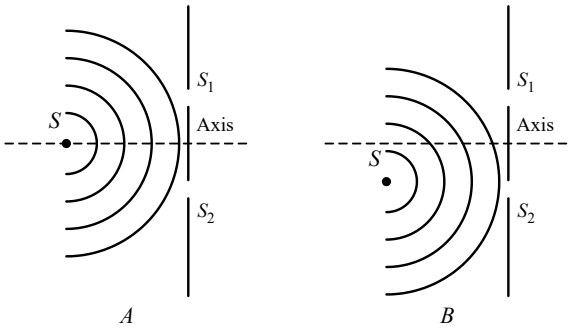
16. (a) State **two** conditions to obtain interference pattern. [2]

Answer

The sources must be coherent: same frequency (wavelength) and a constant phase difference	1
The waves should have (nearly) equal amplitudes and overlap (and be unpolarised or polarised in the same plane)	1
	2

Explanation: Two separate lamps are not coherent, which is why a single source and two slits are used.

(b) A light source, S, is placed at the axis of two slits, S_1 and S_2. The source is then moved away from the axis as shown in the diagram below.



(i) In both cases A and B, the S_1 and S_2 act as coherent sources. State the condition for interference pattern formed. [2]

Answer

A: bright fringes where the path difference $S_2P - S_1P = n\lambda$ (central fringe on the axis) 1

B: the source is off the axis, so the paths from S to the slits differ too: $(SS_2 + S_2P) - (SS_1 + S_1P) = n\lambda$; the pattern shifts (the fringe spacing is unchanged) 1

2

Explanation: Moving the source changes the phase difference between S_1 and S_2, which shifts the whole pattern.

(ii) In A, a monochromatic light source of wavelength 590 nm is used. The distance between S_1 and S_2 is 0.55 mm. If the screen is placed 1.2 m from S_1 and S_2, interference fringes are observed. Calculate the distance of the 4th bright fringe from the central maximum. [3]

Answer

$y_n = \frac{n\lambda D}{d}$ with $n = 4$ 1

$y_4 = \frac{4(590 \times 10^{-9})(1.2)}{0.55 \times 10^{-3}}$ 1

$y_4 = 5.15 \times 10^{-3}$ m 1

3

Total for question **7**

Explanation: The 4th bright fringe is four fringe separations from the central maximum.

17. (a) Using Bohr's model for angular momentum, estimate the speed of an electron moving around the nucleus of nth orbit that is allowed in a hydrogen atom for state $n = 3$. The radius of the third orbit is 4.77×10^{-10} m. [3]

Answer

Bohr: $mvr = \frac{nh}{2\pi}$ with $n = 3$ — 1

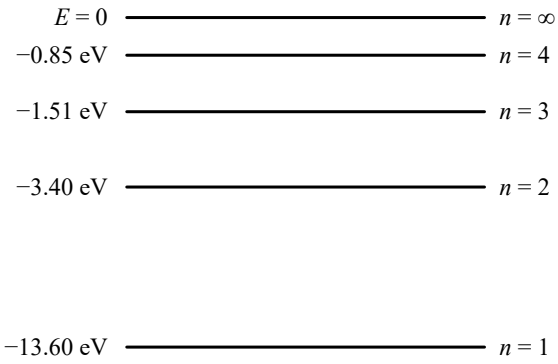
$v = \frac{3(6.63 \times 10^{-34})}{2\pi(9.11 \times 10^{-31})(4.77 \times 10^{-10})}$ — 1

$v = 7.28 \times 10^{5}$ m s^{-1} — 1

3

Explanation: The speed in the nth orbit is proportional to $1/n$.

(b) The energy levels of a hydrogen atom at various states is shown in the diagram below.



(i) What would happen to the photons of energies 12.0 eV and 10.2 eV when they pass through a stable state atomic hydrogen vapour? [2]

Answer

12.0 eV photons pass through unabsorbed: no transition from $n = 1$ has exactly 12.0 eV (the gaps are 10.2, 12.09, 12.75 eV …) — 1

10.2 eV photons are absorbed, exciting electrons from $n = 1$ to $n = 2$ — 1

2

Explanation: A photon is absorbed only if its energy exactly matches the energy difference between two levels.

(ii) If an excited electron experiences a transition from $n = 3$ to $n = 1$, determine the wavelength of the photons that are emitted. [3]

Answer

$\Delta E = -1.51 - (-13.60) = 12.09$ eV — 1

$\lambda = \frac{hc}{\Delta E} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{(12.09)(1.60 \times 10^{-19})}$ — 1

$\lambda = 1.03 \times 10^{-7}$ m — 1

3

Total for question **8**

Explanation: This ultraviolet line belongs to the Lyman series.

Section C [30 marks]

18. (a) Describe the propagation of sound waves in a piece of steel bar. [2]

Answer

Sound travels through the steel as a longitudinal wave: atoms vibrate about their mean positions parallel to the direction of propagation — 1

forming compressions and rarefactions; energy passes from atom to atom through the bonds (much faster than in air) — 1

2

Explanation: The speed of sound in a solid is $\sqrt{(\frac{E}{\rho})}$.

(b) Distinguish between the intensity of sound and the intensity level of sound. [2]

Answer

Intensity: the sound energy per second (power) per unit area perpendicular to the direction of propagation, unit W m^{-2} — 1

Intensity level: a logarithmic measure compared with the threshold of hearing, $\beta = 10 \log (I/I_0)$, in decibels — 1

2

Explanation: The decibel scale matches the way the ear responds to loudness.

(c) An audio system produces a sound of intensity level 80.0 dB at a distance of 55.0 m.
[Threshold of hearing is 1.0×10^{-12} W m^{-2}.]

(i) Determine the intensity of sound at the distance. [2]

Answer

$80.0 = 10 \log \frac{I}{1.0 \times 10^{-12}}$ 1

$I = 1.0 \times 10^{-4}$ W m^{-2} 1

2

Explanation: $I = I_0 \times 10^{\beta/10}$.

(ii) Calculate the intensity level of the sound heard at a distance of 100 m from the source. [2]

Answer

$I_2 = (1.0 \times 10^{-4})(\frac{55.0}{100})^2 = 3.03 \times 10^{-5}$ W m^{-2} 1

$\beta_2 = 74.8$ dB 1

2

Explanation: Assuming the system acts as a point source.

(d) A bird is flying towards a bird-watcher while emitting a sound of frequency 1275 Hz. The bird-watcher hears a sound of frequency of 1350 Hz.
[Speed of sound is 343 m s^{-1}.]

(i) Determine the speed of the bird. [3]

Answer

$f' = f\frac{v}{v - v_s}$ 1

$1350 = 1275\frac{343}{343 - v_s}$ 1

$v_s = 19$ m s^{-1} 1

3

Explanation: The bird (source) moves towards the stationary bird-watcher.

(ii) Explain why there is a difference in the frequency heard by the bird-watcher. [2]

Answer

The bird moves towards the watcher, so successive wavefronts are emitted closer together (shorter wavelength in front of the bird) — 1

More wavefronts reach the watcher per second, so the frequency heard is higher (Doppler effect) — 1

2

Explanation: The speed of sound is unchanged; only the wavelength changes.

(iii) If another bird sits on a tree near the bird-watcher and chips at a frequency of 1270 Hz, calculate the beat frequency heard by the bird-watcher. [2]

Answer

Beat frequency = $|f_1 - f_2| = 1350 - 1270$ — 1

= 80 Hz — 1

2

Total for question **15**

Explanation: Use the frequency actually heard from the flying bird.

19. (a) Write the lensmaker's equation. [2]

Answer

$\frac{1}{f} = (n-1)(\frac{1}{r_1} + \frac{1}{r_2})$ — 1

n = refractive index of the lens material (relative to the surroundings); r_1, r_2 = radii of curvature (positive for convex surfaces) — 1

2

Explanation: State the sign convention used.

(b) A lens with power 4.00 D in air is made from a glass with refractive index of 1.56. The radius of curvature for convex front surface of the lens is 20.0 cm.

(i) Determine the focal length of the lens. [2]

Answer

$f = \frac{1}{P} = \frac{1}{4.00}$ — 1

f = 0.250 m = 25.0 cm — 1

2

Explanation: Power in dioptres is $1/f$ with f in metres.

(ii) Determine the radius and the shape of the other surface of the lens. [3]

Answer

$\frac{1}{25.0} = 0.56(\frac{1}{20.0} + \frac{1}{r_2})$ 1

$r_2 = +46.7$ cm 1

Positive, so the other surface is convex: a biconvex lens 1

3

Explanation: With the convention that convex surfaces have positive radii, a positive result means convex.

(iii) Explain the change in the power of the lens if it is immersed in seawater with a refractive index that is higher than that of the glass. [2]

Answer

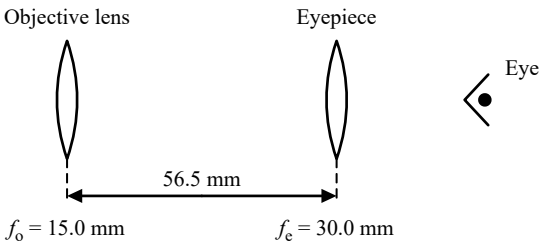
In seawater the relative refractive index n_{glass}/n_{sea} is less than 1, so $(n - 1)$ becomes negative 1

The power becomes negative: the lens becomes a diverging lens 1

2

Explanation: A lens converges light only if it is optically denser than its surroundings.

(c) A microscope that consists of an objective lens and eyepiece is shown in the diagram below.



The focal length of the objective lens and eyepiece are 15.0 mm and 30.0 mm respectively. The lenses are 56.5 mm apart. A sample of microorganism is placed 30.0 mm from the objective lens.

(i) Determine the position of the final image formed. [4]

Answer

Objective: $\frac{1}{v_o} = \frac{1}{15.0} - \frac{1}{30.0} \Longrightarrow v_o = 30.0$ mm behind the objective — 1

Object distance for the eyepiece: $u_e = 56.5 - 30.0 = 26.5$ mm — 1

$\frac{1}{v_e} = \frac{1}{30.0} - \frac{1}{26.5}$ — 1

$v_e = -227$ mm: the final image is virtual, 227 mm from the eyepiece on the objective side — 1

4

Explanation: The intermediate image lies inside the focal length of the eyepiece, which acts as a magnifying glass.

(ii) Calculate the magnification of the microscope. [2]

Answer

$M = m_o \times m_e = \frac{30.0}{30.0} \times \frac{227}{26.5}$ — 1

$M = 8.57$ — 1

2

Total for question **15**

Explanation: The total magnification is the product of the magnifications of the two lenses.

20. (a) State the function of moderator and control rod in a nuclear fission reactor. [2]

Answer

Moderator: slows down fast neutrons so that they can cause further fission of U-235 — 1

Control rods: absorb excess neutrons to control the rate of the chain reaction — 1

2

Explanation: Examples: graphite or water (moderator); boron or cadmium (control rods).

(b) A nuclear reactor undergoes nuclear fission reaction represented by ${}^{235}_{92}\mathrm{U} + {}^{1}_{0}n \rightarrow {}^{127}_{50}\mathrm{Sn} + {}^{104}_{a}\mathrm{Mo} + b\ {}^{1}_{0}n + \mathrm{Q}$.

[Mass of ^{235}U nucleus is 235.0439 u, mass of proton is 1.0073 u, mass of neutron is 1.0087 u, and 1 u ≡ 931.5 MeV.]

(i) Determine the values of a and b. [2]

Answer

Proton number: $92 = 50 + a \Longrightarrow a = 42$ 1

Nucleon number: $235 + 1 = 127 + 104 + b \Longrightarrow b = 5$ 1

2

Explanation: Both proton number and nucleon number are conserved.

(ii) Determine the binding energy per nucleon for ^{235}U in MeV. [3]

Answer

$\Delta m = 92(1.0073) + 143(1.0087) - 235.0439 = 1.8718$ u 1

$E_B = (1.8718)(931.5) = 1743.6$ MeV 1

Per nucleon $= \frac{1743.6}{235} = 7.42$ MeV 1

3

Explanation: U-235 has 92 protons and 235 − 92 = 143 neutrons.

(iii) Given the binding energy per nucleon for ^{127}Sn and ^{104}Mo are 8.248 MeV and 8.349 MeV respectively. Explain why energy is released in the reaction. [2]

Answer

The products have a higher binding energy per nucleon (8.248 and 8.349 MeV) than U-235 (7.42 MeV) 1

so the products are more tightly bound; the increase in total binding energy is released 1

2

Explanation: Equivalently, the products have less total mass than the reactants.

(iv) Calculate the energy released in the reaction in Joule. [2]

Answer

$Q = 127(8.248) + 104(8.349) - 235(7.42) = 172$ MeV 1

$Q = (172 \times 10^6)(1.60 \times 10^{-19}) = 2.76 \times 10^{-11}$ J 1

2

Explanation: Free neutrons have no binding energy.

(v) If the electricity output power of the reactor is 100 MW and the efficiency of the reactor is 20%, calculate the mass of ^{235}U used in 30 days. Assume the nuclear reactor operates 24 hours per day. [4]

Answer

Energy released in 30 days = $\frac{(100 \times 10^6)(30 \times 24 \times 3600)}{0.20} = 1.30 \times 10^{15}$ J — 1

Number of fissions = $\frac{1.30 \times 10^{15}}{2.76 \times 10^{-11}} = 4.70 \times 10^{25}$ — 1

Mass = $(4.70 \times 10^{25})(235.0439)(1.66 \times 10^{-27}\ \text{kg})$ — 1

Mass = 18.4 kg — 1

4

Total for question **15**

Explanation: With 20% efficiency, the reactor must release five times the electrical energy delivered.